

Nicholas P. Dana, PhD

Entrepreneur, Collaborator, AI & Cross-Industry Innovator

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SUMMARY (Updated June 2026)

Innovative data scientist and AI engineer integrating advanced analytics and AI to accelerate scientific discovery and product innovation. Proven leader in building scalable systems, automating complex pipelines, and translating R&D breakthroughs into operational tools that drive value across business applications and technology sectors.

EXPERIENCE

Elephas Biosciences — *Principle AI Scientist & Developer*

June 2021 – Present | Remote / Madison, WI

- Architected and deployed **multi-modal analytics pipelines** in cloud-integrated ML workflows leveraging image, cytokine, and assay data to predict drug response in *ex vivo* tumor models
- **Leadership Impact:** Technical lead for SME group (8) developing custom analytics achieving **>96% training data accuracy** and **87% concordance** with validation samples
- **Strategic Initiative:** Prototyped analytic methods leading to company's first Lab Derived Test (LDT) offering (publication in review)
- **Strategic Initiative:** Led group efforts to develop and implement a testing framework (adapted TOST approach) to validate LDT method transfer to multiple international sites
- Co-authored multiple conference & BioRxiv publications and abstracts advancing translational oncology (see below)

GeoVisual Analytics — *Computer Vision & Machine Learning Scientist*

2018 – 2021 | Boulder, CO

- Built and trained multiple **deep-learning CV models** for agricultural analytics using drone-based imagery
- Developed **best-in-class models** for crop maturation and yield forecasting, leveraging multi-modal data from custom image systems, weather sensors and feeds, and geolocation data
- **Quantifiable Result:** Achieved **0-day median error** in harvest date prediction and **<5% error** in crop yield estimation across multiple production sites throughout central California and Arizona

Essen BioScience (Sartorius) — *Algorithm Engineer*

2016 – 2018 | Ann Arbor, MI

- Designed and implemented segmentation and denoising algorithms enabling automated live-cell image analysis with minimal user input
- **Quantifiable Result:** Developed a computer vision library for time-series imaging, achieving **99.5% data fidelity** at **14x probabilistic compression efficiency**

CERTIFICATIONS & CREDENTIALS

Post Graduate Program in AI Agents for Business Applications

University of Texas at Austin - McCombs School of Business, May 2026

Accelerated course giving students the technical and conceptual depth to design, deploy, monitor, improve and govern Agentic AI solutions for a variety of business applications

EDUCATION

Doctor of Philosophy, Biomedical Engineering

The University of Texas at Austin

Dissertation: *Photoacoustic Imaging for Tissue Characterization and Image Guidance in Cardiovascular Applications*

Bachelor of Science, Physics

University of Utah

Focus in Medical Physics, Mathematics and Chemistry

SELECTED PUBLICATIONS

- Dana N. *et al.*, “**Prediction of immunotherapy response using live tumor fragments from routine clinical biopsies**” Manuscript in review (link)
- Ramasubramanian T.S. *et al.*, “**A live tumor fragment platform to assess immunotherapy response in core needle biopsies,**”, *Journal of Translational Medicine* 2025 (link)
- Dana N. *et al.*, “**Development and characterization of cytokine-based classifiers to predict response to immunotherapy using an ex vivo live tumor fragment platform**”, *Society of Immunotherapy in Cancer 2025 Poster* (link)
- Dana N. *et al.*, “**Optimization of dual-wavelength intravascular photoacoustic imaging of atherosclerotic plaques using Monte Carlo optical modeling**”, *Journal of Biomedical Optics*, 2017 (link)
- Dana N. *et al.*, “**Overview of Photoacoustic Imaging**”, *Advances in Medical Physics*, 2016 edition.

TECHNICAL SKILLS

Tools: Python, SQL, C++, C#, R, MATLAB, Docker

AI / Data Science: GenAI, AgenticAI, LangChain / LangGraph, PyTorch / Tensorflow, Deep Learning, Scikit, Pandas, Bayesian Methods, Dimensionality Reduction, Unsupervised Learning, Computer Vision

Domain Expertise: Optical Imaging, Assay Development and Validation, Multi-modal Analysis, High-dimensional Analysis, Bioinformatics, Computational Modeling